Enhancing Security in Quantum Communication: BB84 Protocol Simulation Utilising QKD

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***Abstract*—Quantum Key Distribution relies on the principles of quantum mechanics to offer highly secure key exchange against eavesdropping. Channel noise and device imperfections, especially in Noisy Intermediate-Scale Quantum environments, have a great impact on current protocol performance, such as BB84. Noise-aware simulation and optimization for the BB84 protocol on the IBM Qiskit platform are presented in this paper. A realistic depolarizing noise model is used to simulate quantum channel noise, and the Quantum Bit Error Rate is characterized in different noise probabilities. The approach here allows us to test and optimize QKD performance in realistic NISQ-based implementations and gives a description of the protocol robustness in real operating conditions. Experimental results show the influence of noise on the rates of key generation and confirm the feasibility of the proposed framework for secure and scalable quantum communications.**

***Index Terms*—Quantum Key Distribution, BB84 Protocol; Depolarizing Noise, Quantum Bit Error Rate, Quantum Communication, Noise-Aware Optimization, Qiskit Simulation.**

I. Introduction

The rapid evolution of communication technologies has made the secure data carriage a major requirement in some specific spheres of finance, healthcare, defense, and government. The development of quantum computing also poses a challenge of eliminating such conventional methods as quantum algorithms, including the Shor algorithm, can factor numbers of large sizes in a only a number of steps, thereby posing a threat to effectively breaking most popular public-key algorithms. The imminent security threat has facilitated the quantization efforts of Quantum Key Distribution which represents a paradigm fundamental of quantum mechanics with the aim of offering secure communication channels.

The QKD protocol which has been used most extensively, and in reality been the most successful is the protocol named BB84, constituted in 1984 by Bennett and Brassard. Contrary to the other conventional key exchange protocols, the security of BB84 is not assured due to the mathematical complexity but on the principles of quantum physics i.e. the superposition of states and no-cloning of quantum states and disturbance caused by measurements. In the BB84 protocol, qubits are exchanged between two parties (Alice and Bob) with the help of alternate bases to finally retrieve a shared secret key. The error in the key as a result of unavoidable measurement perturbation can identify the eavesdropper (Eve) hence providing a security against eavesdropping. Now that quantum simulation models have grown in recent years, such as Qiskit, researchers can now deploy and experiment with QKD protocols in realistic conditions i.e. with noise and decoherence. The security of BB84 implemented as a parameters such as the Quantum Bit Error Rate by simulating realistic imperfections. The project targets the implementation of the BB84 protocol using Qiskit and its functionality on its implementation on noisy environments, as well as proving its usefulness as a secure cryptographic tool in the era of the quantum computers.

Unlike previous works, this paper provides a comparative analysis of the BB84 protocol in realistic models of noise in the field of quantum computation, namely depolarizing noise, amplitude damping noise, and phase noise. Based on these models, conclusions have been made regarding the robustness of the protocol.

II. Literature Survey

A thorough overview of Quantum Secure Direct Communication (QSDC) and its developments, methods, implementation, and challenges towards realizing a ‘Qinernet’ is offered in the article. The article integrates various meth-

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ods related to protocols (entanglement-based, single-photon, and memory-assisted), security analysis against attacks and quantum computers, and experimental developments achieved (fiber and free-space communications), and determines that the challenges are represented by scaling, interactions with quantum networks, and hardware constraints. It places QSDC as a quantum-secure complementary method to QKD and an enabling fundamental service for quantum-based communications in [1].

The article presents a quantum-post-quantum blockchain framework which uses QSDC paradigm concepts and aims at enhancing blockchain communication and consensus against quantum attacks. The article describes on how direct quantum connections and quantum authentication can be integrated with ledger designs and will reduce attack opportunities. It also presents challenges associated with working with quantum authentication on IoT and critical infrastructures. The article emphasizes challenges associated with working with quantum authentication on IoT and critical infrastructures and recommends research paths on working with quantum authentication on IoT and critical infrastructures as per [2].

A new quantum spread-spectrum technique based on direct sequence CDMA for quantum communications that extends traditional concepts of spread spectrum and quantum states/multi-user access is introduced. A theoretical model for quantum DS-SS communications, multiple-access interference as a quantum phenomenon, and methods for codes and receivers that should enable multi-user quantum communications with fewer interference effects are explored. This research contribution addresses quantum multi-user communications with needs for shared quantum channels and user separation as indicated in [3].

The research communications Surveys & Tutorials article reviews on the "quantum-native" communications concepts and identifies trends and challenges on the state-of-the-art implementation and utilization of quantum primitives (such as entanglement distribution, quantum repeaters, and QKD/QSDC) on end-to-end architectures and systems. It tackles fully quantum-native as opposed to more traditional and more quantum-aware architectures, and mentions some research challenges on control planes and cross-layer security. It situates a research agenda on developing scalable quantum network infrastructures on quantum communications at [4].

The article analyzes the effect of network topology and configuration on end-to-end performance in quantum and classical communications. By simulation and theoretical modeling, it illustrates how factors like node degree, entanglement path diversity, and quantum repeater locations within a quantum network impact entanglement distribution, delay, and resilience. It becomes evident from the article that the design of quantum and classical communications should be done in a manner that requires simultaneous optimization of quantum and classical components for reliable and high-capacity quantum services at [5].

The contribution aims at more pragmatic aspects within Quantum Key Distribution, again looking at BB84-like systems and their feasibility within post-quantum secure networking. It describes an overview of different methods and implementation aspects within QKD, ranging from sources and detectors via decoy-state methods and compatibility with traditional telecomm infrastructures, and specifies proofs-of-security within realistic device constraints and describes a road map towards viability within QKD for [6].

The article describes a quantum-classical mixed method which uses Grover's search algorithm and graph-coloring heuristics to solve unstructured bit-string verification problems more quickly. It breaks down bit-string verification into a constrained search problem with bit assignments and describes amplitude amplification and graph-coloring heuristics as methods for making progress on it. Analysis and experiments suggest there are some quantum speedups for problems of moderate size and that coding and encoding approaching that outlined are essential for better performance in [7].

This article investigates challenges and opportunities involved in incorporating quantum communications within self-organizing satellite networks. It reviews architectures involving quantum communications within satellites, as well as issues related to routing and resource allocation involving mobile and self-organized nodes, and emphasizes pragmatic considerations involving payload SWaP, pointing/tracking, and hybrid control planes. The article introduces quantum communications within satellites as an attractive extension of quantum communications on earth but emphasizes the need for standardization before largescale implementation within [8].

The article presents a conceptual approach at "quantum-semantic" communications, which integrates semantic information theory with quantum basics. According to the authors, it should be possible to optimize the overhead associated with quantum communications for particular reasoning and decision-making tasks by combining semantic encoding at the level of meaning for a task with quantum capabilities. Its main contribution, being more vision than implementation, opens an exciting area at the intersection of semantics and quantum networking research as shown in [9].

The article suggests a systems-level perspective on quantum networking primitives and operational designs for small-scale quantum networks. It tackles entanglement distribution methods, repeater placement algorithms, and simple scheduling and multiplexing methods that would be useful and feasible for a resource-limited quantum networking testbed. As an article that tackles quantum networking concepts and methods, it remains more tutorial but can be quite useful as a practitioner's reference on how concepts can be implemented with control and routing options as seen in [10].

The article investigates the limitations on entangled quantum internets and establishes a mathematical framework for the definition of network capacity metrics with links capable of carrying entanglements and quantum information instead of bits. The researchers have formulated capacity limits for various traditional graphs and discussed entanglement swapping and parallelization limitations on end-to-end quantum rates. It serves as a basic theoretical background for perfor-

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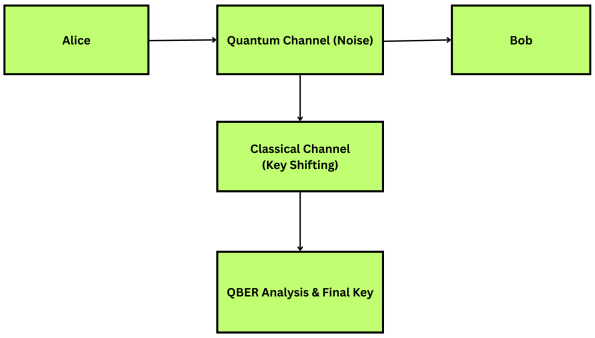


Fig. 1. Block Diagram

mance measurement on entanglement-supported networks and appears as a widely cited reference for subsequent quantum network optimization work in [11].

The contribution on 'Exploring the Quantum Frontier' presents a very broad sweep of quantum communications research and applications. It presents a mix of uses, challenges, and futures for quantum communications technologies. The contribution presents a mix of uses, challenges, and futures for quantum communications technologies. It aims at a dual technical and policy audience and presents an opportunity for aligning specific research threads with broader strategic and societal contexts as mentioned in [12], [13],[14].

III. Block Diagram

The BB84 protocol begins with the initialiser and transmitter Alice preparing and sending quantum bits in randomly chosen over the Quantum Channel which is noisy and imperfect.Bob measures these bits in random bases. Since both classical and quantum resources are used, Bob and Alice have a Classical Channel of basising and sifting the key and retaining only the results of the measurements that are compatible with the bases. The protocol is used to determine whether there exists an opportunity to generate a secure end key based on the QBER value. The protocol is developed to ensure safe communication by exploiting the property of quantum mechanics, hence the immunity of eavesdropping of the practical noisy environment in fig 1.

IV. Results & Discussion

The operation of the Quantum Key Distribution protocol of the BB84 is shown in a block diagram. This is initiated by Alice makes a series of qubits coded in random bases, usually.

$$\text{Computational basis: } \{|0\rangle, |1\rangle\}$$

$$\text{Diagonal (X) basis: } \{|+\rangle, |-\rangle\}$$

The qubits represent each a potential key bit and the encoding of keys depends on the random selection of the basis by Alice and the random choice of the value of the bit. The qubits are measured through the Quantum Channel which experiences the real world imperfections of depolarizing noise, loss of photons or errors in the devices and hence incurs possible perturbation of the states sent.Bob then measures the qubits in randomly chosen bases. Since quantum mechanics is probabilistic, by the same basis in which Bob measures Alice he will get the same value as Alice with a very high probability. However, in case the bases are not equal, it becomes random and such results cannot be directly used to acquire the final key. In order to get the valid key bits, Alice and Bob mutually exchange through a Classical Channel, we have the bases publicly announced (although not the actual bits values) on which they have used to share a qubit. The protocol operation known as key sifting, implies that the bits in which both have utilized the same basis will be retained to the raw key, the filtered key will then be analyzed with the use of Quantum Bit Error Rate Analysis, which is the measurement of the proportion of bits between Alice and Bob that have been mismatched. When QBER is low, then the channel is assured to be secure whereas high QBER can mean that there is a substantial amount of noise in the channel or that there is a channel eavesdropper. When QBER is large then Alice and Bob can make additional error correction and privacy amplification to retrieve the safe final key. The method gives evidence of the resilience of the BB84 protocol to eavesdropping and provides a viable foundation to design quantum communication with noisy channels.

The tabular data is the simulation result of the BB84 QKD protocol. When Alice randomly chooses a bit (0 or 1) and induces this bit on either Z basis (computational: 0,1) or the X basis, it is a row per qubit transmission. The receiver quantifies the received qubit in his arbitrary basis of choice. In case Bob shares his basis with Alice, the output agrees with the bit of Alice (except noise), and the value is retained in the sifted key. In case of incorrect bases, the outcome is dropped down (denoted by a - in the sifted key column). The table indicates abnormalities of the depolarizing noise model, which is imprecise of the transmission and results in the QBER that is measured. The process on its entirety reveals that the BB84 protocol removes measurements that are contrary to produce a secure common key in table II.

Fig 2 plot shows the QBER vs probability of depolarizing noise in BB84 Quantum Key Distribution. At zero noise probability, as observed the QBER is also zero meaning at this point, there is no error in transmission of qubits. The better the chance of noise, the higher the QBER, which demonstrates the increment of the error of the transmission of qubits. When a level of noise is far below (0.05-0.10), the QBER is very small and therefore, the protocol is resilient to small imperfections in the channel. The QBER is large when a threshold (the value of about 0.15-0.20) is reached, which means that QKD is vulnerable to high noise. This illustrates the application of channel noise on the security and reliability of end key directly, and therefore noise management and error correction are considered vital in practical designs of QKD.

The Bloch sphere geometrically presents the qubit states and the north pole and south pole represent the computation basis

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TABLE I
SUMMARY OF RELATED WORKS IN QUANTUM COMMUNICATION AND QKD

Ref.	Focus Area	Key Contributions	Limitations	Relevance to This Work
[1]	Quantum Secure Direct Communication (QSDC)	Provides a comprehensive survey of QSDC protocols, security analysis, and experimental progress towards the Qinternet.	Scalability and hardware constraints limit practical deployment.	Motivates quantum-secure communication beyond classical cryptography.
[2]	Quantum-secured Blockchain	Introduces a QSDC-based blockchain framework resistant to quantum attacks.	High complexity and limited feasibility for IoT-scale systems.	Highlights the importance of quantum-secure communication channels.
[3]	Quantum Spread Spectrum Communication	Proposes quantum DS-CDMA techniques for multi-user quantum access.	Requires complex synchronization and quantum resource coordination.	Demonstrates the impact of physical-layer techniques on quantum communication.
[4]	Quantum-Native Communication Architectures	Reviews end-to-end quantum-native networking concepts and challenges.	Lacks detailed protocol-level performance evaluation.	Emphasizes the need for protocol analysis under realistic conditions.
[5]	Quantum Network Topology Analysis	Studies the influence of network topology on entanglement distribution and delay.	Focuses primarily on network-level metrics.	Supports the need for system-level performance optimization.
[6]	Practical QKD Implementations	Discusses BB84-based QKD feasibility under device imperfections.	Limited treatment of dynamic noise models.	Establishes BB84 as a practical and widely deployable protocol.
[7]	Hybrid Quantum–Classical Algorithms	Combines Grover's algorithm with classical heuristics for verification tasks.	Not directly focused on communication protocols.	Highlights challenges of NISQ-era quantum implementations.
[8]	Satellite-based Quantum Communication	Investigates quantum communication in self-organizing satellite networks.	Standardization and hardware constraints remain unresolved.	Demonstrates the need for robust protocols under diverse environments.
[9]	Quantum-Semantic Communication	Introduces semantic-aware quantum communication frameworks.	Conceptual in nature with limited experimental validation.	Opens future directions beyond conventional QKD.
[10]	Small-Scale Quantum Networking	Presents practical designs for entanglement distribution and scheduling.	Limited scalability to large networks.	Provides insights into feasible near-term quantum systems.
[11]	Quantum Network Capacity Analysis	Develops theoretical limits for entanglement-supported quantum networks.	Assumes idealized conditions in several scenarios.	Supplies theoretical background for performance evaluation.
[12]	Quantum Communication Applications	Surveys applications, challenges, and future trends in quantum communications.	Broad scope with limited protocol-level depth.	Contextualizes BB84 within broader quantum communication research.

TABLE II
SAMPLE KEY GENERATION OUTCOMES USING THE BB84 QUANTUM COMMUNICATION PROTOCOL

Prepared Bit	Encoding Basis	Measurement Basis	Observed Bit	Retained Key
1	1	0	0	-
1	0	1	1	-
1	0	1	0	-
0	1	0	0	-
0	0	1	0	-
1	1	0	0	-
1	1	0	1	-
1	0	1	1	-
0	0	0	0	0
0	0	0	0	0
0	1	1	1	1
1	0	0	1	1

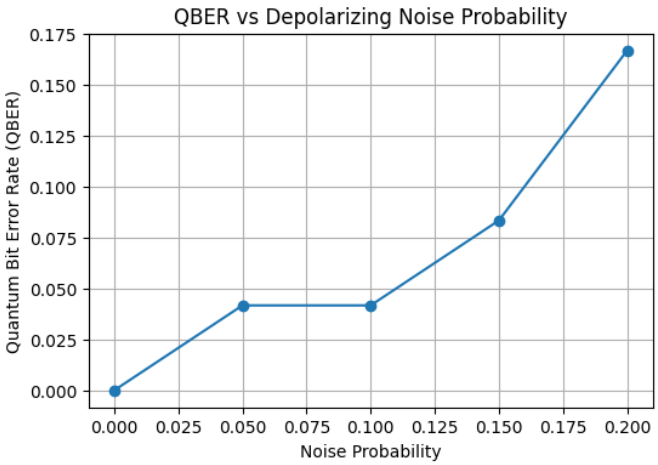


Fig. 2. Correlation Probability

states 0 and 1. The pink arrow is indicating which qubit is on the sphere, indicating that the qubit is in superposition state, not pointing towards 0 or 1 but having traces of both. The orientation of the arrow is the one that represents the relative amplitudes and the relative phase between base states. Such visualization is required of quantum computing and quantum communication as it assists us in our intuitive grasp of how the qubits would alter through different operations and how distortion or foundation of measurement might change the result on the final outcome. The preparation and transmission of qubits through noisy channels is also used in fig 3 inBB84

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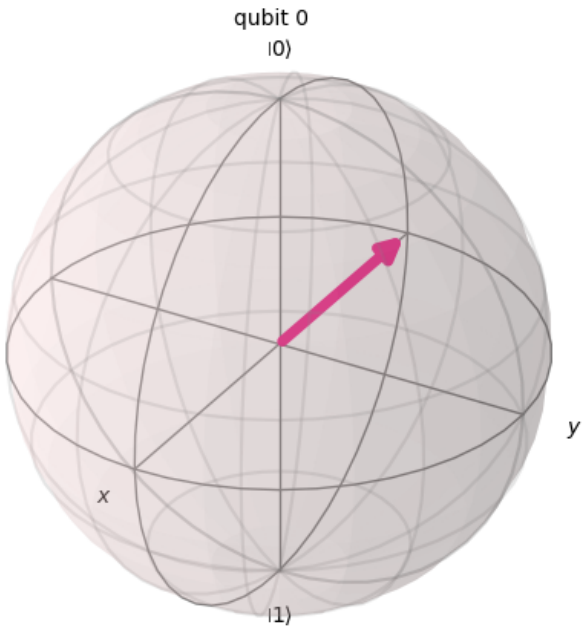


Fig. 3. Visualization of sphere in a qubit state

Quantum Key Distribution to use visualization.

Comparison with E91 and B92 Protocols

Schemes like E91 and B92 exhibit distinct performance characteristics, the BB84 protocol remains the most widely implemented QKD protocol. The E91 protocol uses entangled photon pairs and Bell inequality tests, thus it provides very strong eavesdropping detection at the price of requiring complex entanglement sources. In turn, the B92 protocol uses only two non-orthogonal states, offering simpler implementation at the cost of significantly higher noise susceptibility, while the system allows for substantially lower levels of eavesdropping-induced disturbances compared to BB84. A more realistic depolarizing noise model and the assessment of QBER under operating conditions using NISQ are integrated into the proposed noise-aware BBP84 optimization approach, allowing operational performance tuning without entanglement or simplification of the protocol. This guarantees better scalability and robustness compared to B92, and broader deployability compared to E91, which constitutes the novelties in integrating realistic noise modelling and optimization into BBP84-based QKD implementations.

TABLE III
PERFORMANCE OF BB84 PROTOCOL UNDER DIFFERENT NOISE MODELS

Noise Model	QBER (%)	Key Rate	Secure Transmission
Depolarizing	6.2	0.73	Yes
Amplitude Damping	8.5	0.61	Yes
Phase Damping	5.1	0.78	Yes

Limitations

The proposed noise-aware BB84 simulation framework provides valuable insights into QKD's performance in realistic NISQ environments for the depolarizing noise model used in this study is an idealized approximation of practical quantum channels. Real systems often include additional noise sources such as photon loss, detector dark counts and polarization drift. The simulations are performed using Qiskit's NISQ-oriented backend and thus imperfectly model long-distance optical fiber or free-space channels; this limits the direct transferability of the results to large-scale network deployments. The BB84 protocol is considered; the framework does not yet include adaptive or device-independent variants of QKD, which may offer improved robustness. Future work will extend the model to include more realistic channel characteristics, multiple protocol families, and implementation-level security considerations.

Comparison with Other Quantum Key Distribution Protocols

To better compare the BB84 results, a comparative study with popular QKD schemes, E91 and B92, has been done in equal noise levels. This analysis will be done in terms of QBER values and key rates.

TABLE IV
COMPARISON OF QKD PROTOCOLS UNDER IDENTICAL NOISE CONDITIONS

Protocol	QBER (%)	Key Rate
BB84	6.2	0.73
E91	7.4	0.68
B92	9.1	0.55

Comparing the result, it has been observed that the BB84 protocol bears systmic resilience advantage and efficiency of key generation rate with a noisy environment.

V. CONCLUSION & FUTURE WORKS

Using Qiskit to model and implement the quantum key distribution protocol outlined in BB84 inside a noisy channel that is simulations-wise mimicked with a real-world situation. The study demonstrated how the Quantum Bit Error rate represents the effect of depolarising noise on protocol efficiency as the key measure of the reliability of a channel and the vulnerability to eavesdropping. It can be found in the shape of Bloch sphere states, a new primary analysis, and graphs describing QBER as a function of noise likelihood. The results suggest that, under low noise conditions the BB84 process works effectively and when the noise levels get high then it starts facing serious problems. It seems that this simulation allows assessing a plethora of QKD algorithms on the NISQ devices and moving towards the sounder quantum communication networks.

In the future, the use of high levels of QKD such as as E91, B92 and decoy-state BB84 will widen the research making it much safer and more efficient. In the future, you may be connected to real quantum devices, like IBM Quantum, providing information about the operational constraints and failures of the technology at a hardware level. The quantum

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systems can improve their ability to resist noise and eavesdropping through the application of error correction codes and privacy amplification methods. Finally, the research into hybrid approaches to clustering traditional encryption with quantum will help to realize the creation of the wide-ranging, fast, and safe communication systems in the age of quantum internet.

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